

FoodBridge Campus

AI Food Waste Donation Matcher

Product Requirements Document (PRD)

FoodBridge Campus is a campus-based AI food rescue platform where donors can post extra food, receivers can claim available meals, and volunteers can prioritize urgent pickups. The system reduces food waste, supports students in need, and tracks real social impact through an easy-to-use dashboard.

Version	1.0
Prepared for	Student AI Portfolio / Hackathon MVP
Recommended MVP Stack	Next.js + Tailwind CSS + Supabase + OpenAI/Gemini API + Vercel
Main Users	Donors, Receivers, Volunteers, Admins

1. Product Overview

FoodBridge Campus is a web-based platform that connects people or organizations with extra food to people who need food. It is designed mainly for university campuses, where leftover food often comes from cafes, food courts, events, student clubs, bakeries, restaurants near campus, and individual students.

Instead of letting edible food go to waste, donors can post available food on the platform. Students, campus volunteers, or NGOs can view the available food and claim it before it expires. The platform also uses simple AI to suggest urgency level, estimate pickup priority, and help users write clearer food descriptions.

2. One-Sentence Pitch

FoodBridge Campus is an AI-powered food rescue platform that helps cafes, events, and students donate extra food to nearby students or NGOs before it goes to waste.

3. Problem Statement

Food waste is a serious problem, especially on university campuses. After events, meetings, cafe closing hours, or student activities, extra food is often thrown away even though it is still safe to eat. At the same time, some students face financial pressure and may struggle to afford regular meals.

The current process is usually unorganized:

- Donors do not know who needs the food.
- Students do not know where extra food is available.
- Food may expire before someone collects it.
- There is no proper claiming system.
- NGOs or volunteers cannot easily coordinate pickup.
- Donors may not know how urgent the food is.
- There is no record of how much food was saved.

FoodBridge Campus solves this by creating a simple donation-matching system where extra food can be posted, claimed, collected, and tracked.

4. Product Goals

1. Reduce food waste on campus.
2. Help students or communities access free leftover food.
3. Make food donation simple and fast.
4. Help cafes and event organizers donate food responsibly.
5. Allow NGOs or student volunteers to coordinate food pickup.
6. Use AI to estimate urgency and prevent food from expiring.
7. Track social impact such as meals saved and food rescued.
8. Build a simple but impressive social-impact AI project.

5. Target Users

Primary Users

User Group	Examples	Main Need
Food Donors	Campus cafes, food court vendors, student event organizers, clubs, restaurants, bakeries, individual students	Post extra food quickly and responsibly
Food Receivers	Students, student volunteers, welfare groups, NGOs, low-income student communities	Find and claim available food clearly

Admin / Campus Coordinator	University admin, student representative council, welfare department, sustainability club, NGO coordinator	Monitor platform activity and safety
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6. User Personas

Persona	Role	Goal	Pain Points	Need
Mr. Hafiz	Campus cafe owner	Donate unsold food before closing	No time to contact many people; needs quick pickup; wants proof that food was claimed	A simple form to post available food quickly
Aina	University student	Find free meals near campus	Limited food budget; does not know when free food is available; cannot travel far	A listing page showing food, pickup location, and expiry time
Thaqif	Club event committee	Donate leftover food after an event	Many extra food packs; food may spoil soon; needs quick claiming	Fast donation post and claim system
Sarah	Food rescue volunteer	Collect and redistribute food	Needs quantity, urgency, contact, and pickup time	Dashboard showing high-priority food donations

7. Core Features

Feature 1: User Registration and Login

Users can create accounts and log in based on their role. Roles include Donor, Receiver, Volunteer/NGO, and Admin. Required fields include full name, email, password, phone number, role, optional organization name, and campus/location.

Priority: High

Feature 2: Donor Food Posting Form

Donors can post extra food available for donation. Inputs include food name, category, quantity, pickup location, pickup deadline, expiry time, preparation time, optional image, halal status, vegetarian status, allergy information, contact number, and notes.

Priority: Very High

Feature 3: AI Food Description Helper

AI helps donors improve food post descriptions by making them clearer, detecting missing details, suggesting allergy warnings, creating pickup instructions, and generating short listing titles.

Priority: Medium

Feature 4: AI Urgency Scoring

The system calculates how urgent a donation is using current time, expiry time, pickup deadline, food type, quantity, preparation time, and storage condition. Urgency levels are Low, Medium, High, and Critical.

Priority: Very High

Feature 5: Food Listing Page

Receivers browse available food donations with image, food name, quantity, location, expiry time, urgency level, halal/vegetarian tags, allergy warning, donor name, and claim button.

Priority: Very High

Feature 6: Food Claim System

Receivers can claim food, enter quantity, confirm pickup time, and submit a claim request. Claim statuses include Pending, Approved, Rejected, Collected, Expired, and Cancelled.

Priority: Very High

Feature 7: Donor Claim Management

Donors can view claim requests, approve or reject claims, update remaining quantity, mark food as collected, cancel listings, and contact the receiver.

Priority: High

Feature 8: Pickup Confirmation

After pickup, donor or receiver confirms that the food was collected. For MVP, this can be a manual button. Future versions can use QR code, OTP, or photo proof.

Priority: Medium to High

Feature 9: Impact Dashboard

The dashboard shows total food posts, meals saved, claims completed, donors, receivers helped, estimated food waste reduced, top donors, and high-impact events.

Priority: Medium

Feature 10: Admin Dashboard

Admin can monitor food listings, users, claims, impact analytics, expired posts, and flagged posts. Admin can remove inappropriate posts.

Priority: Medium

Feature 11: Notification System

Users receive in-app notifications when new food is available, a claim is requested, a claim is approved, or critical pickup is needed.

Priority: Medium

Feature 12: AI Donation Summary

AI generates a short impact summary for donors or event organizers after a successful donation.

Priority: Low to Medium

8. MVP Scope

MVP Features

9. Landing page
10. Login/signup
11. User role selection
12. Donor food posting form
13. Food listing page
14. Claim button

15. Claim status tracking
16. AI urgency level
17. AI description helper
18. Basic impact dashboard

Not in MVP

- Real map integration
- GPS distance tracking
- Payment system
- Real-time chat
- QR code pickup confirmation
- Mobile app
- Machine learning prediction model
- Advanced NGO routing
- Weather-based food safety analysis
- Automatic WhatsApp notification
- Food safety certification system

9. User Flow

Flow 1: Donor Posts Extra Food

19. Donor logs in.
20. Donor clicks Post Food.
21. Donor fills in food details.
22. AI checks missing information.
23. AI generates urgency level.
24. Donor publishes listing.
25. Food appears on listing page.
26. Receiver claims food.
27. Donor approves claim.
28. Receiver collects food.
29. Donor marks it as collected.
30. Impact dashboard updates.

Flow 2: Receiver Claims Food

31. Receiver logs in.
32. Receiver opens food listing page.
33. Receiver filters by location or urgency.
34. Receiver selects a food post.
35. Receiver checks expiry time and allergy info.
36. Receiver clicks Claim.
37. Receiver enters quantity and pickup time.
38. Donor approves claim.
39. Receiver collects food.
40. Receiver marks as completed.

Flow 3: Volunteer Finds Urgent Donation

41. Volunteer logs in.
42. Volunteer opens urgent pickup dashboard.
43. System shows High or Critical urgency food.
44. Volunteer claims pickup.
45. Volunteer collects food from donor.
46. Volunteer redistributes food.
47. Volunteer marks task completed.

Flow 4: Admin Monitors Platform

- 48. Admin logs in.
- 49. Admin views all food posts.
- 50. Admin checks expired or flagged posts.
- 51. Admin removes invalid listings.
- 52. Admin views monthly impact report.

10. Functional Requirements

ID	Requirement	Priority
FR1	User can register and log in	High
FR2	User can select role as donor, receiver, volunteer, or admin	High
FR3	Donor can create food donation listing	Very High
FR4	Donor can upload food image	Medium
FR5	System can calculate urgency level	Very High
FR6	AI can improve food description	Medium
FR7	Receiver can view available food listings	Very High
FR8	Receiver can filter food by category, urgency, and halal status	High
FR9	Receiver can claim food	Very High
FR10	Donor can approve or reject claim	High
FR11	Donor can update remaining quantity	High
FR12	User can mark food as collected	High
FR13	System can update listing status automatically	High
FR14	System can hide expired food	High
FR15	Admin can view all users and listings	Medium
FR16	Admin can remove inappropriate posts	Medium
FR17	System can show total meals saved	Medium
FR18	System can show impact dashboard	Medium
FR19	System can send basic in-app notifications	Medium
FR20	AI can generate donation impact summary	Low

11. Non-Functional Requirements

Category	Requirement
Usability	Interface must be simple and fast to use
Mobile Responsiveness	Must work well on phone and laptop
Performance	Food listing page should load quickly
Security	User data must be protected
Privacy	Contact details should only be shown after claim approval
Reliability	Claim status must update correctly
Safety	Food expiry and allergy information must be clearly shown
Transparency	AI urgency score must explain its reason
Accessibility	Text should be readable with clear buttons
Scalability	System should support more campuses later

12. AI System Design

AI Component 1: Description Cleaner

Purpose: Improve donor-written food descriptions. Input includes food name, quantity, location, expiry time, and notes. Output includes a clear food description, allergy reminder, and pickup instruction.

AI Component 2: Urgency Explainer

Purpose: Explain why a listing is urgent. Input includes food type, quantity, current time, expiry time, and pickup deadline. Output includes urgency level and a short reason.

AI Component 3: Impact Summary Generator

Purpose: Generate a short positive impact message after a successful donation. Input includes meals saved, food type, receiver type, and pickup time.

13. Urgency Scoring Logic

For MVP, use rule-based logic instead of complex machine learning.

Time Remaining = Expiry Time - Current Time

Time Remaining	Urgency
More than 6 hours	Low
3 to 6 hours	Medium
1 to 3 hours	High
Less than 1 hour	Critical
Already expired	Expired

Increase urgency by one level if quantity is more than 20 meals, the food is cooked rice or meat-based, the pickup deadline is less than 2 hours, food is not refrigerated, or the location is far from main campus.

Example:

Current Time: 6:00 PM

Expiry Time: 8:00 PM

Quantity: 25 packs

Food Type: Fried rice

Base Urgency: High

Quantity Factor: +1

Final Urgency: Critical

14. Food Status Logic

Status	Meaning
Available	Food can still be claimed
Partially Claimed	Some quantity has been claimed
Fully Claimed	All food has been claimed
Collected	Food has been collected
Expired	Food expired
Cancelled	Donor cancelled listing

15. Suggested Tech Stack

Area	Recommended Technology
Frontend	React / Next.js
Styling	Tailwind CSS
Backend	Next.js API Routes / Node.js

Database	Supabase
Authentication	Supabase Auth
File Storage	Supabase Storage
AI	OpenAI API / Gemini API
Maps, later	Google Maps API / Mapbox
Charts	Recharts
Deployment	Vercel
Form Validation	Zod
Icons	Lucide React

Recommended student stack: Next.js + Tailwind CSS + Supabase + OpenAI/Gemini API + Vercel

16. Database Structure

Users Table

Field	Type	Description
user_id	UUID	Unique user ID
full_name	Text	User full name
email	Text	User email
phone	Text	Contact number
role	Text	Donor, receiver, volunteer, admin
organization_name	Text	Optional organization
campus_location	Text	User campus area
created_at	Timestamp	Account creation time

Food Listings Table

Field	Type	Description
food_id	UUID	Unique food listing ID
donor_id	UUID	User ID of donor
food_name	Text	Name of food
category	Text	Food category
description	Text	Food description
quantity_total	Number	Original quantity
quantity_available	Number	Remaining quantity
pickup_location	Text	Collection location
pickup_deadline	Timestamp	Latest pickup time
expiry_time	Timestamp	Food expiry time
prepared_time	Timestamp	Food preparation time
halal_status	Boolean	Whether food is halal
vegetarian_status	Boolean	Whether food is vegetarian
allergy_info	Text	Allergy warning
storage_condition	Text	Room temp, refrigerated, heated
image_url	Text	Food image
urgency_level	Text	Low, Medium, High, Critical
status	Text	Available, claimed, collected, expired
created_at	Timestamp	Listing creation time

Claims Table

Field	Type	Description
claim_id	UUID	Unique claim ID
food_id	UUID	Food listing ID
receiver_id	UUID	User ID of receiver
quantity_claimed	Number	Quantity requested
pickup_time	Timestamp	Planned pickup time
claim_status	Text	Pending, approved, rejected, collected
receiver_note	Text	Optional note

created_at	Timestamp	Claim creation time
updated_at	Timestamp	Last update

Notifications Table

Field	Type	Description
notification_id	UUID	Unique notification ID
user_id	UUID	Receiver of notification
title	Text	Notification title
message	Text	Notification message
is_read	Boolean	Read status
created_at	Timestamp	Creation time

Impact Table

Field	Type	Description
impact_id	UUID	Unique impact record
food_id	UUID	Related food listing
donor_id	UUID	Donor user ID
receiver_id	UUID	Receiver user ID
meals_saved	Number	Number of meals rescued
completed_at	Timestamp	Collection completion time

17. Key Screens

Screen	Purpose / Components
Landing Page	Explain the project and attract users. Includes hero, problem statement, how it works, roles, impact statistics, and CTA buttons. Hero text: Rescue Extra Food. Feed Students. Reduce Waste.
Login / Sign Up Page	Allow users to access the platform with email and password. Sign up collects name, email, password, phone, role, and campus location.
Dashboard	Different dashboard by role. Donor sees listings and claims. Receiver sees available and claimed food. Volunteer sees urgent pickups. Admin sees users, listings, claims, and meals saved.
Post Food Page	Donors enter food name, category, quantity, location, expiry time, pickup deadline, halal/vegetarian status, allergy info, image, AI improve description, and publish.
Food Listings Page	Receivers browse food cards with search, filters, urgency badges, and claim buttons.
Food Detail Page	Shows food image, name, quantity, donor, location, expiry, urgency explanation, allergy info, claim form, and safety note.
Claim Management Page	Donor reviews pending claims, receiver name, quantity requested, pickup time, approve/reject, and mark as collected.
Impact Dashboard	Shows meals saved, completed donations, top donors, monthly trend, estimated waste reduced, and AI-generated impact summary.

18. UI Design Direction

The design should feel clean, friendly, modern, trustworthy, student-focused, and social-impact oriented.

Purpose	Color Idea
Primary	Green
Secondary	Cream / light beige

Accent	Orange
Alert	Red
Neutral	White / gray

Use a modern dashboard style with rounded cards, large readable buttons, simple icons, clear status badges, and a mobile-first layout.

19. Safety and Trust Requirements

FoodBridge must be careful because food safety matters.

Required Safety Notice

Please check the food condition before consuming. FoodBridge Campus only connects donors and receivers and does not guarantee food safety.

Donor Responsibility

- Confirm food is still edible.
- Confirm food is not spoiled.
- Confirm food was stored properly.
- Provide allergy information if known.
- Estimate expiry time honestly.

Receiver Responsibility

- Understand allergy risks.
- Collect before expiry time.
- Inspect food before consuming.

20. Success Metrics

Metric	Target
Number of food listings created	100 in pilot phase
Completed claims	70% of listings
Expired unclaimed food	Less than 25%
Average claim time	Under 30 minutes
Number of meals saved	500 in pilot phase
Active donors	20+
Active receivers	100+
User satisfaction	4/5
Donor repeat usage	40%
Volunteer pickup completion	80%

21. Risks and Limitations

Risk	Problem	Solution
Food Safety Issues	Food may be spoiled or unsafe.	Show safety disclaimer, require expiry time and allergy info, let users report unsafe food, and allow admin removal.
Fake or Inaccurate Listings	Users may post fake food listings.	Require login, allow reporting, use admin moderation, and add verified donor badges later.
Food Not Collected in Time	Food may expire before pickup.	Use urgency score, pickup deadline, expired status, critical highlighting, and volunteer notifications.
Too Many Claims for Same Food	Multiple users may claim the same	Track available quantity, lock quantity

	food.	after claim, show remaining quantity, and prevent over-claiming.
Privacy Concerns	Donor and receiver contact details may be exposed.	Show phone number only after claim approval, use in-app status updates, and hide sensitive data from public listing.

22. MVP Build Plan

Phase	Build
Phase 1: Basic Web App	Landing page, login/signup, role selection, basic dashboard
Phase 2: Donor Food Posting	Food posting form, image upload, save listing to database, display donor active listings
Phase 3: Listing and Claim System	Public food listing page, food detail page, claim form, claim status, donor approval/rejection
Phase 4: AI Features	AI description cleaner, AI urgency explanation, rule-based urgency score, AI impact summary
Phase 5: Impact Dashboard and Polish	Meals saved counter, completed donation count, clean UI, mobile responsive design, demo data, README, demo video

23. Minimal Prototype Version

Pages

- 53. Landing page
- 54. Donor post food page
- 55. Food listing page
- 56. Food detail page
- 57. Claim confirmation page
- 58. Impact dashboard

Features

- Add food post
- View food cards
- Claim food
- Urgency badge
- Expiry countdown
- Meals saved counter
- AI generated description

24. Example Demo Scenario

Food: Fried rice
Quantity: 30 packs
Location: Dewan Seminar Block A
Pickup Deadline: 9:00 PM
Expiry Time: 10:00 PM
Halal: Yes
Allergy: Contains egg

AI Output:
Urgency: Critical
Reason: 30 packs of cooked food expire in less than 2 hours.
Recommended pickup: Before 9:00 PM.

Receiver sees the listing, claims 5 packs, donor approves, receiver collects, and the impact dashboard updates to show 5 meals saved.

25. Example AI Prompts

Prompt 1: Food Description Cleaner

You are FoodBridge AI, an assistant for a campus food rescue platform.

Rewrite this food donation listing to be clear, short, and helpful.

Food name: {{food_name}}
Quantity: {{quantity}}
Pickup location: {{pickup_location}}
Pickup deadline: {{pickup_deadline}}
Expiry time: {{expiry_time}}
Halal status: {{halal_status}}
Vegetarian status: {{vegetarian_status}}
Allergy information: {{allergy_info}}
Storage condition: {{storage_condition}}
Extra notes: {{notes}}

Return:

1. Listing title
2. Short description
3. Pickup instruction
4. Allergy reminder
5. Safety reminder

Keep the tone friendly and professional.

Prompt 2: Urgency Explanation

You are FoodBridge AI, an urgency assistant for food rescue.

Analyze this food donation:

Food type: {{food_type}}
Quantity: {{quantity}}
Current time: {{current_time}}
Expiry time: {{expiry_time}}
Pickup deadline: {{pickup_deadline}}
Storage condition: {{storage_condition}}

Classify urgency as one of: Low, Medium, High, Critical, or Expired.

Explain:

1. The urgency level
2. The reason
3. Recommended pickup time

Keep the answer under 60 words.

Prompt 3: Impact Summary

You are FoodBridge AI.

Write a short impact summary for a completed food donation.

Food donated: {{food_name}}
Meals saved: {{meals_saved}}
Donor type: {{donor_type}}
Receiver type: {{receiver_type}}

Pickup completed at: {{completed_time}}

Keep it under 50 words. Make it encouraging and professional.

26. Example README Description

FoodBridge Campus is an AI-powered food waste donation matcher that helps cafes, students, and event organizers donate extra food before it goes to waste.

The platform allows donors to post available food, receivers to claim food, and volunteers to prioritize urgent pickups. AI is used to generate clearer food descriptions, classify urgency levels, and summarize social impact.

Built with Next.js, Tailwind CSS, Supabase, and AI integration.

27. Competitive Advantage

Normal Food Donation Method	FoodBridge Campus
WhatsApp group messages	Organized food listing system
No urgency level	AI urgency score
No clear claim tracking	Claim and approval flow
No impact tracking	Meals saved dashboard
Food may expire unnoticed	Expiry countdown and priority
Hard to manage many claims	Quantity and status tracking

28. Why This Project Is Strong for Portfolio

- Real social impact
- Full-stack development
- AI integration
- Database design
- User role system
- Dashboard design
- Problem-solving ability
- Sustainability awareness
- Campus/community relevance

It is realistic to build without complex hardware, sensors, or expensive APIs.

29. Recommended Final MVP Name Options

- 59. FoodBridge Campus
- 60. MealBridge AI
- 61. CampusBite Rescue
- 62. WasteLess Campus
- 63. ResQMeal
- 64. ShareBite
- 65. LeftoverLink
- 66. CampusFeast
- 67. PlateForward
- 68. BiteBack AI

Best name: FoodBridge Campus. It sounds professional, clear, and social-impact focused.

30. Final Product Summary

FoodBridge Campus is a campus-based AI food rescue platform where donors can post extra food, receivers can claim available meals, and volunteers can prioritize urgent pickups. The system reduces food waste, supports students in need, and tracks real social impact through an easy-to-use dashboard.